

A Prospective Randomized Comparison of Bone–Patellar Tendon–Bone and Hamstring Grafts for Anterior Cruciate Ligament Reconstruction

Gauti Laxdal, M.D., Jüri Kartus, M.D., Ph.D., Leif Hansson, M.D., Mats Heidvall, M.D., Lars Ejerhed, M.D., Ph.D., and Jon Karlsson, M.D., Ph.D.

Purpose: The aim of this study was to compare the results after arthroscopic anterior cruciate ligament (ACL) reconstruction using central-third, bone–patellar tendon–bone (BPTB group), 3-strand semitendinosus (ST group), or 4-strand semitendinosus/gracilis (ST/G group) autografts. **Type of Study:** Prospective randomized trial. **Methods:** A randomized series of 134 patients, all with unilateral ACL rupture was included in the study. In all 3 groups, interference screw fixation of the graft was used at both ends and 125 of 134 (93%) of the patients returned for the follow-up examination after 26 months (range, 20 to 43 months). The preoperative assessments in all 3 groups were similar in terms of gender, Tegner activity level, Lysholm score, KT-1000 measurements, 1-leg hop test, and the knee-walking test. **Results:** At follow-up, the knee-walking test was significantly worse in the BPTB group than in the ST group ($P = .0004$) and ST/G group ($P < .0001$). Furthermore, the knee-walking test was significantly worse at follow-up than preoperatively in the BPTB group ($P < .0001$). The corresponding findings were not made in the other 2 groups. A significant reduction in knee laxity and an increase in activity level compared with the preoperative assessments were found in all 3 groups, without any significant differences between the groups. **Conclusions:** Two years after ACL reconstruction, the use of ST and ST/G autografts rendered significantly less discomfort during the knee-walking test than the use of BPTB autografts. However, in terms of functional outcome and knee laxity, the groups displayed no significant differences. **Level of Evidence:** Level I. **Key Words:** Anterior cruciate ligament—Patellar tendon—Semitendinosus/gracilis—Autograft—Randomized.

There is still some controversy as to which type of graft is best for the reconstruction of the anterior cruciate ligament (ACL).¹ Two types of grafts are used most often, either an autogenous central-third bone–patellar tendon–bone (BPTB) graft^{2–4} or an au-

togenous graft involving the hamstring tendons, usually a quadruple semitendinosus/gracilis (ST/G) graft.^{5–7} Using the BPTB graft for open or arthroscopic reconstruction of the ruptured ACL has been a standard procedure in recent decades and it is sometimes referred to as the gold standard.⁸ The results have been reported as good and reproducible in several studies.^{9–11} However, there are obvious problems when it comes to using this method, especially in terms of donor-site morbidity.^{4,12,13} Anterior knee pain, disturbances in knee sensitivity, and kneeling discomfort are also commonly encountered problems. There is also an obvious need for an alternative graft option in the event of revision surgery.

During the last decade, the hamstring graft in the form of 3- or 4-strand ST or 4-strand ST/G has been used increasingly as an alternative for the BPTB graft.

From the Departments of Orthopaedics, Sahlgrenska University Hospital (G.L., J. Karlsson), Göteborg; NÄL-Hospital (J. Kartus, L.E.), Trollhättan; Arthro Clinic (L.H.), Stockholm; and Lövenströmska Hospital (M.H.), Stockholm, Sweden.

Supported by grants from the Research and Development Department at the Northern Älvsborg/Bohus County Council and the Swedish National Centre for Research in Sports.

Address correspondence and reprint requests to Jüri Kartus, M.D., Ph.D., Department of Orthopaedics, NÄL Hospital, SE-461 85 Trollhättan, Sweden. E-mail: juri.kartus@vgregion.se

© 2005 by the Arthroscopy Association of North America

0749-8063/05/2101-3803\$30.00/0

doi:10.1016/j.arthro.2004.09.014

However, there are still only a few controlled studies in which these options are compared.^{1,5-7,14-18} In studies comparing BPTB and hamstring tendon grafts, the functional outcome parameters have been similar, but the patients operated on using hamstring tendon grafts have had significantly less donor-site and anterior knee pain problems in some studies.^{6,7,16,17} The principal consideration in the choice of grafts has traditionally been strength. A large BPTB graft or a multiply looped hamstring graft must be used initially to approach the strength of the normal ACL.¹⁹ It was recently reported that a double-looped hamstring graft was inferior in terms of residual knee laxity compared with a traditional BPTB graft after 3 to 7 years of follow-up.¹ However, the subjective knee function was found to be similar.

The aim of this prospective randomized study was to compare the outcome of ACL reconstruction after using a BPTB graft, a 3-strand ST graft, or a 4-strand ST/G graft. The hypotheses were that in the short term, the use of ST or ST/G grafts in ACL reconstruction would cause less donor-site morbidity in terms of subjective anterior knee pain and would result in better knee-walking ability. Furthermore, that the use of ST and ST/G grafts would reduce knee laxity and render good functional outcome to the same extent as the use of BPTB grafts does.

METHODS

Patients

Patients with a symptomatic unilateral chronic ACL rupture from the catchment area of three Swedish medical centers were prospectively randomized for reconstruction using an ipsilateral BPTB graft, an ipsilateral 3-strand ST graft, or an ipsilateral 4-strand ST/G graft. The randomization was performed using closed envelopes. The initial aim was to randomize 150 patients to 3 different treatment groups, but, owing to the time factor, the study was terminated after randomizing 134 patients.

Surgical Technique

Six senior surgeons, all experienced in using the 3 techniques, performed the reconstructions. In the BPTB group, the arthroscopic transtibial technique²⁰ and interference screw fixation²¹ were used during the index procedure. The central third of the patellar tendon was harvested through a 60- to 80-mm long vertical incision. The tendon defect was left open and the defects in the patella and the

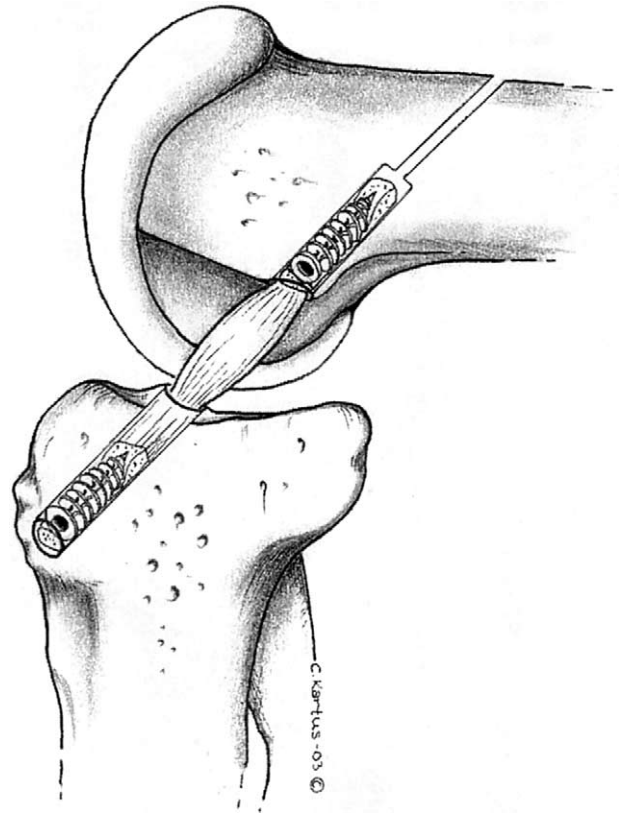


FIGURE 1. In the BPTB group, a 7-mm and a 9-mm sharp threaded interference screw was used on the femoral and tibial side, respectively. (Copyright 2003, © Catarina Kartus. Used with permission.)

proximal tibia were not bone grafted because this has previously been shown not to decrease postoperative anterior knee problems.^{22,23} The proximal bone block was sized to 9 mm and the distal bone block to 10 mm. The bone tunnels were prepared in a standard transtibial fashion. A 7-mm and a 9-mm Acuflex "silk" interference screw (Acufex Microsurgical, Mansfield, MA) were used on the femoral and tibial side, respectively (Fig 1).

In the ST group, the graft was harvested through a 3-cm oblique incision over the pes anserinus. The tendons were palpated and the sartorius fascia was incised parallel to the fibers of the fascia, just above the thicker and more distally inserted semitendinosus tendon. After the vinculae had been cut under visual control, the tendon was harvested with a semiblunt, semicircular open tendon stripper (Acufex). The tendon was constructed to a 3-strand graft measuring a minimum of 7 cm in length. Two No. 5 nonresorbable Ticron sutures (Sherwood Medical, St. Louis, MO)

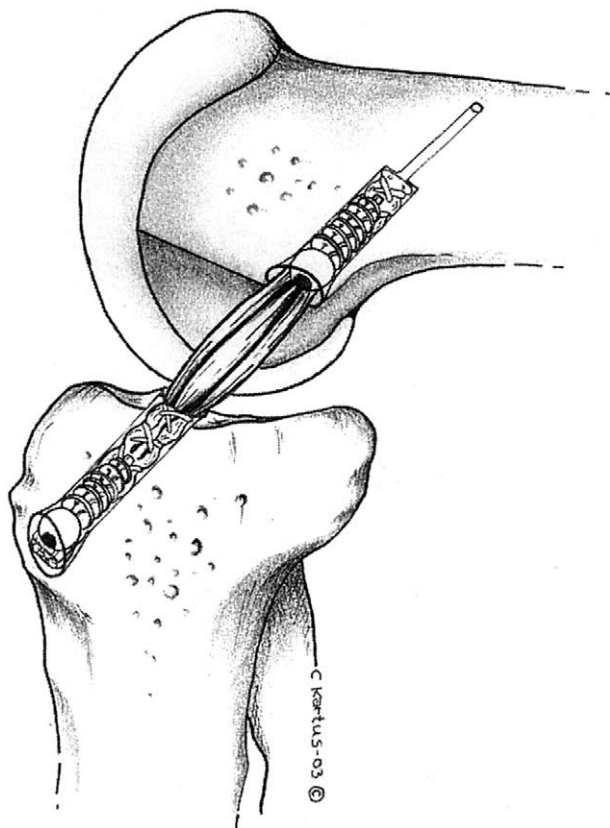


FIGURE 2. In the ST and ST/G groups, a 7-mm, round-headed, soft-threaded interference screw was used on both the femoral and tibial sides. (Copyright 2003, © Catarina Kartus. Used with permission.)

were used as the lead sutures at the distal and proximal ends. Resorbable No. 1 Vicryl (Ethicon Inc, Somerville, NJ) sutures were used for the modified baseball stitches at the distal and proximal ends of the ST graft. The bone tunnels were drilled using specially designed Stepped Router drills (Smith & Nephew, Andover, MA). The femoral tunnel was drilled through a medial portal with the knee in maximum flexion. A 7-mm soft-threaded, round-headed RCI interference screw (Smith & Nephew) was used on both the femoral and tibial side (Fig 2).²⁴

In the ST/G group, both the semitendinosus and gracilis tendons were harvested and constructed to a 4-strand graft. Otherwise, the surgical technique was identical to that used in the ST group. After the femoral screw had been inserted, firm traction was applied to the graft during the insertion of the tibial screw, with the knee in hyperextension, in all 3 groups.

Clinical Assessments and Follow-up

Three independent physiotherapists, who were not involved in the rehabilitation, performed all the preoperative and postoperative assessments. The KT-1000 arthrometer (MEDmetric, San Diego, CA) was used for the assessment of knee laxity.²⁵ The Lysholm score,²⁶ Tegner activity level,²⁶ and the International Knee Documentation Committee (IKDC)²⁷ evaluation system were used to evaluate the functional outcome. The Lysholm score was self-administered according to Höher et al.²⁸

Range of motion was measured to the nearest 5°, using a goniometer. Any loss of motion of 5° or more in extension, including hyperextension in comparison with the normal contralateral knee, was categorically classified as loss of motion. The corresponding measurement was performed in terms of flexion.

Measurements of disturbances in sensitivity in the anterior knee region were performed by palpation and measured in square centimeters.²⁹ The patients were categorically classified as having subjective anterior knee pain if they registered pain during stair walking, sitting with the knee in 90° of flexion, and during or after activity.

The classification of kneeling discomfort was based on the knee-walking test. This test involves direct loading of the anterior knee region while walking forward on a hard floor on the knees without any protection such as clothing or kneepads. The test is subjectively classified by the patient as "OK, unpleasant, difficult, or impossible" to perform.^{4,29,30} The 1-leg hop test³¹ was used to evaluate the functional performance.

Rehabilitation

All patients underwent rehabilitation according to the same protocol, permitting immediate full weight-bearing and full range of motion, including full hyperextension using the weight of the operated leg.³² However, an external load was not used during the first 6 postoperative weeks from 30° to full (hyper)extension. No rehabilitation brace was used in the study.³³⁻³⁵ Closed-chain exercises were started immediately postoperatively. Running was permitted at 3 months and contact sports at 6 months at the earliest, provided that the patient had regained full functional stability.

Statistical Methods

Median and range values are presented. In terms of both continuous and noncontinuous variables, the

TABLE 1. Preoperative Data for Patients Who Attended the Follow-up

	BPTB Group	ST Group	ST/G Group	Significance
No. of patients	40	39	39	
Age (yr)	28 (16-52)	24 (12-41)	26 (15-35)	NS ($P = .17$)
Gender, F/M	11/29	17/22	11/28	NS ($P = .24$)
Preinjury Tegner activity level	8 (5-10)	9 (4-9)	9 (5-9)	NS ($P = .97$)
Time between injury and index operation (mo)	14.5 (2-240)	14 (2-360)	9 (2-194)	NS ($P = .73$)
Follow-up period (mo)	25.5 (23-33)	27 (20-36)	26 (24-43)	NS ($P = .11$)

NOTE. There were no significant differences between the groups in terms of demographics.

Kruskal-Wallis test was first used to test if there was any significant difference between the 3 groups. If the Kruskal-Wallis test revealed a significant difference, the groups were tested pair-wise with each other using the Mann-Whitney U test. The Wilcoxon signed-rank test was used for comparisons of the preoperative and postoperative data within the groups. The χ -square test was first used to compare the dichotomous variables between the 3 groups. If the χ -square test revealed a significant difference, the groups were tested pair-wise with each other using the Fisher exact test; $P < .05$ was considered statistically significant.

Ethics

The Human Ethics Committee at the Medical Faculty at the Göteborg and Stockholm Universities approved the study. All the patients gave their informed consent.

RESULTS

The study comprised 134 patients. Follow-up was performed on 125 of 134 patients (93%); 9 patients were lost to follow-up. Of the patients who attended the follow-up, 5 of 125 underwent revision ACL surgery during the follow-up period. One patient in the ST/G group and 1 in the ST group suffered early septic failure of the graft and 3 patients (1 from each group) reruptured their ACL after a return to sports.

Furthermore, 1 patient from the BPTB group ruptured his contralateral ACL during the follow-up period and 1 patient from the ST/G group had an old contralateral ACL reconstruction, which was missed during the randomization procedure. All these patients ($n = 7$) were excluded. This left 118 patients; 40 in the BPTB group, 39 in the ST group, and 39 in the ST/G group, on whom the comparisons between the treatment groups were based. The groups were comparable in terms of age, gender, preinjury Tegner activity

level, time between the injury and index operation, and length of follow-up period (Table 1).

Meniscal injuries that were treated before the index operation, at the index operation, or during the follow-up period were registered in 23 of 40 patients in the BPTB group, 23 of 39 in the ST-group, and 23 of 39 in the ST/G group (not significant).

At follow-up, there were no significant differences in terms of either the Lysholm score or the Tegner activity level between the groups. However, all 3 groups improved significantly between the preoperative assessment and follow-up (Table 2). At follow-up, 58% in the BPTB group, 59% in the ST group, and 69% in the ST/G group had returned to a Tegner activity level of 6 or above (not significant). In terms of the 1-leg hop test, at follow-up, the ST/G-group was significantly better than both the BPTB and the ST groups (Table 2).

The KT-1000 anterior and total side-to-side difference at 89 N revealed no significant differences between the study groups either preoperatively or at follow-up. However, both the KT-1000 anterior and total side-to-side differences displayed a significant decrease in all 3 groups between the preoperative assessment and follow-up (Table 3). Stratifying the postoperative KT-1000 anterior data did not reveal any significant differences between the 3 study groups (Table 3). The final IKDC evaluation at follow-up did not reveal any significant difference between the study groups (Table 4).

The disturbance in anterior knee sensitivity was a median of 51.5 cm² (range, 0 to 832 cm²) (8 missing) in the BPTB group, 56 cm² (0 to 364 cm²) (6 missing) in the ST group, and 30 cm² (0 to 330 cm²) (8 missing) in the ST/G group (not significant) ($P = .73$). The knee-walking ability was significantly better in both the ST and the ST/G group compared with the BPTB group (Table 5). In the BPTB group, the knee-walking ability was significantly worse at follow-up than pre-

TABLE 2. *Lysholm Score, Tegner Activity Level, and 1-Leg Hop Test Preoperatively and at Follow-up*

	BPTB Group	ST Group	ST/G Group	Significance
Lysholm score				
Preop (1 missing value)	74 (49-99) points	68 (21-100) points	75 (36-99) points	NS ($P = .09$)
Follow-up (6 missing values)	94 (22-100) points	89 (49-100) points	95 (39-100) points	NS ($P = .06$)
Significance preop v postop	$P < .0001$	$P < .0001$	$P < .0001$	
Tegner activity level				
Preop	4 (1-9)	4 (0-8)	4 (2-8)	NS ($P = .91$)
Follow-up (6 missing values)	6 (2-10)	6 (2-9)	7 (1-9)	NS ($P = .33$)
Significance preop v postop	$P = .0009$	$P = .0002$	$P < .0001$	
1-leg hop test				
Preop (10 patients were not able to perform the test)	90 (45-104)%	82 (40-108)%	88 (57-113)%	NS ($P = .07$)
Follow-up (11 patients were not able to perform the test)	94 (52-108)%	93 (39-120)%	99 (79-120)%	$P = .004$
				$P = .006$; ST v ST/G;
				$P = .003$; BTB v ST/G;
				NS ($P = .88$; BPTB v ST)
Significance preop v postop	NS ($P = .14$)	$P = .002$	$P < .0001$	

NOTE. A significant improvement was found in all 3 groups in terms of the Lysholm score and Tegner activity level. The 1-leg hop test was significantly improved in the ST and ST/G groups, but not in the BPTB group.

operatively. The corresponding finding was not made in the ST and ST/G groups (Table 5).

There were no significant differences at follow-up between the study groups in terms of loss of motion compared with the healthy contralateral side, in extension or flexion (Table 6). In the ST and ST/G groups, the subjective anterior knee pain decreased significantly between the preoperative evaluation and the follow-up. The corresponding finding was not made in the BPTB group (Table 7).

DISCUSSION

In this prospective randomized study, we compared the well-established and most frequently used central-third BPTB graft with a 3-strand ST and 4-strand ST/G graft. All 3 graft types rendered similar improvements in knee function and a significant reduction in laxity. At follow-up, the most important, clinically significant difference between the 3 methods was that the ST and the ST/G groups had a better

TABLE 3. *KT-1000 Anterior and Total Side-to-Side Laxity Measurements Preoperatively and at Follow-up*

	BPTB Group	ST Group	ST/G Group	Significance
KT-1000 anterior side-to-side difference at 89 N				
Preop	4 (−4-15) mm	4 (−3-14) mm	4 (−3-16) mm	NS ($P = .78$)
Follow-up (7 missing values)	1 (−5-8) mm	2 (−4-9) mm	1 (−5-8) mm	NS ($P = .06$)
Significance preop v postop	$P < .0001$	$P = .001$	$P < .0001$	
KT-1000 total side-to-side difference at 89 N				
Preop	5 (2.5-15) mm	4 (0-14) mm	4.5 (2.5-16) mm	NS ($P = .98$)
Follow-up (7 missing values)	1 (−6.5-8) mm	2 (−4-9) mm	1 (−5.5-8) mm	NS ($P = .11$)
Significance preop v postop	$P < .0001$	$P = .0002$	$P < .0001$	
Stratified KT-1000 anterior side-to-side difference at 89 N at follow-up (7 missing values)				
<−1 mm	13%	6%	5%	NS ($P = .76$)
−1-3 mm	71%	69%	78%	
>3 mm but <6 mm	11%	19%	14%	
6 mm and more	5%	6%	3%	

NOTE. A significant reduction in anterior and total side-to-side laxity was found in all 3 groups, but there was no significant difference between the groups.

TABLE 4. *Final IKDC Classification at Follow-up*

Final IKDC Classification at Follow-up (6 missing values)	BPTB Group	ST Group	ST/G Group	Significance
Normal	9/38 (24%)	5/36 (14%)	10/38 (26%)	NS ($P = .15$)
Nearly normal	10/38 (26%)	15/36 (42%)	18/38 (47%)	
Abnormal	12/38 (32%)	11/36 (31%)	7/38 (18%)	
Severely abnormal	7/38 (18%)	5/36 (14%)	3/38 (8%)	

NOTE. No significant difference in terms of the final IKDC classification was found between the groups.

ability to knee-walk. One important finding was that there was no significant difference in terms of knee laxity between the groups at the final follow-up. In a recent study, Beynnon et al.¹ compared the BPTB graft with a 2-strand ST/G graft. In their prospective randomized study, they found that the 2 treatments produced comparable results in terms of patient satisfaction, subjective knee function, and activity level. However, the BPTB graft produced superior results in terms of knee laxity, as measured with the KT-1000 laxity meter, and pivot-shift grades. Furthermore, there was a significant difference in terms of knee flexor muscle strength in favor of the BPTB group. In the present study, a significant improvement was found in all 3 groups in terms of the Lysholm score and Tegner activity level without any significant differences between the groups. This indicates that all 3 graft types produce similar and, in overall terms, satisfactory knee function, including a significant increase in activity level, at least after 2 to 3 years of follow-up, as long as the knee-walking ability is not taken into consideration. It should also be kept in mind that the follow-up period in the study by Beynnon et al.¹ was longer than that of the present study. Increased laxity over time has previously been sug-

gested, especially in patients in whom ST/G grafts have been used. This could of course also be the case in our patients.

When it comes to the unanswered questions relating to ACL reconstruction, the choice of graft and the type of graft that is best for a reconstruction remain controversial.^{11,36} Several factors should be considered when a graft is chosen. The most important is probably the initial strength of the graft/fixation complex. The main factor affecting the structural strength during the first weeks after the insertion of the graft is not the initial graft strength itself. The fixation points on the femoral and even more on the tibial side, appear to be the weakest points, as suggested by Shino and Pflaster.³⁷ In the present study, interference screws were used on both the femoral and tibial sides in all patients. This may be questioned, especially in relation to the hamstring grafts, although it does not appear to be a problem in clinical terms. After bone-to-bone or tendon-to-bone healing has taken place, probably between 4 and 12 weeks postoperatively, the attachment site is no longer the weakest link.^{38,39} To minimize the early differences between the groups, we strictly adhered to the same rehabilitation protocol for all 3 groups.

TABLE 5. *Knee-Walking Ability Preoperatively and at Follow-up*

Knee-Walking Test	BPTB Group	ST Group	ST/G Group	Significance
Preop				
Normal	20/40 (50%)	20/39 (51%)	21/39 (54%)	NS ($P = .99$)
Unpleasant	17/40 (42.5%)	16/39 (41%)	14/39 (36%)	
Difficult	3/40 (7.5%)	2/39 (5%)	3/39 (8%)	
Impossible	0/40 (0%)	1/39 (3%)	1/39 (3%)	
Follow-up (6 missing values)				
Normal	6/38 (16%)	17/36 (47%)	25/38 (65%)	$P < .0001$ BTB v ST ($P = .0004$) BTB v ST/G ($P < .0001$) ST v ST/G ($P = .04$)
Unpleasant	11/38 (29%)	10/36 (28%)	11/38 (29%)	
Difficult	4/38 (11%)	6/36 (17%)	2/38 (5%)	
Impossible	17/38 (45%)	3/36 (8%)	0/38 (0%)	
Significance preop v postop	$P < .0001$	NS ($P = .14$)	NS ($P = .22$)	

NOTE. At follow-up, the knee-walking test was difficult or impossible to perform for more than 50% of patients operated on with a BPTB graft. Furthermore, the patients in the BPTB group displayed a significantly worse knee-walking ability at follow-up than preoperatively.

TABLE 6. *Loss of Motion of 5° or More in Extension and Flexion at Follow-up*

Loss of Motion	BPTB Group	ST Group	ST/G Group	Significance
Extension (6 missing values)	13/38 (34%)	8/36 (22%)	6/38 (16%)	NS ($P = .16$)
Flexion (6 missing values)	24/38 (63%)	22/36 (61%)	26/38 (68%)	NS ($P = .79$)

NOTE. No significant differences were found between the groups.

There is some evidence that all tendon tissue will lose some of its initial strength during the early healing period.¹⁰ The grafts never regain their original strength and only recover between 10% and 50% of normal stiffness.⁴⁰ However, we believe that 3- or 4-strand hamstring tendon grafts lose less of both strength and stiffness compared with 2-strand grafts.

In the literature, there are 2 reports of studies involving a meta-analysis comparing the results after ACL reconstruction using BPTB or hamstring-tendon autografts. Both studies reveal less laxity in the BPTB group.^{41,42} There are a few previous prospective randomized studies that compared BPTB grafts and ST/G grafts.^{1,5-7,14-18,43} Some of these studies involved 2-strand ST/G grafts and revealed an increase in side-to-side laxity measurements compared with the BPTB graft.^{1,43} The main difference in recent studies has been less anterior knee pain and donor-site morbidity in favor of the patients who underwent ACL reconstruction using hamstring tendon grafts.^{6,7,16,17} It should be remembered that no long-term randomized studies of ST/G grafts are yet available. It is still unknown whether these grafts will produce functionally stable knees after 10 years or more. In a nonrandomized prospective study with a 5-year follow-up, Pinczewski et al.⁴⁴ concluded that both hamstring and BPTB grafts restored knee stability in a reliable way. Therefore, it appears that modern fixation techniques and 3- or 4-strand hamstring grafts will produce results after ACL reconstruction that are well on a par with the results after using BPTB grafts.

The impact and importance of the knee-walking test

could be questioned. However, this can pose a major problem in young people, who need to kneel on the floor during work and leisure-time activities. This test has been shown to be a reliable and reproducible test for evaluating anterior knee problems after ACL reconstruction in several publications.^{4,29,30,45,46} Furthermore, in a recent publication by Ejerhed et al.,⁶ this variable revealed the only significant difference between BPTB and hamstring tendon autografts in a prospective randomized study. In fact, in the present study, the patients in the BPTB group were worse off in terms of knee-walking ability after surgery than before. Therefore, we are in favor of using ST or ST/G grafts for ACL reconstruction.

In a previous study,⁴ it was shown that the knee-walking ability had an inverse correlation with the loss or disturbance of anterior knee sensitivity after ACL reconstruction using patellar tendon autografts. In the present study, there were no significant differences in the loss of sensitivity between the study groups, which means that all the graft-harvesting techniques jeopardized the infrapatellar branch of the saphenous nerve to the same extent. Because of the location of the skin incisions in the present study, possible subcutaneous neuroma formations must bear weight during knee-walking after BPTB graft harvest but not after ST and ST/G graft harvest. This theory has previously been proposed by Ejerhed et al.⁶ as one of the reasons for the difference in knee-walking ability found in the present study.

The strength of this study is its prospective and randomized design and the fact that unbiased observ-

TABLE 7. *Subjective Anterior Knee Pain Preoperatively and at Follow-up*

Subjective Anterior Knee Pain	BPTB Group	ST Group	ST/G Group	Significance
Preop	14/40 (35%)	19/39 (49%)	14/39 (36%)	NS ($P = .38$)
Follow-up (6 missing values)	11/38 (29%)	9/36 (25%)	2/38 (5%)	$P = .02$ $P = .02$ ST v ST/G NS ($P = .70$) BPTB v ST $P = .007$ BPTB v ST/G
Significance preop v postop	NS ($P = .45$)	$P = .02$	$P = .0005$	

NOTE. The patients in the ST and ST/G groups had a significant decrease in subjective anterior knee pain compared with the preoperative assessment. The corresponding finding was not made in the BPTB group. At follow-up, there was no significant difference between the BPTB and ST groups.

ers, who were not involved in the surgery or rehabilitation, performed the preoperative and postoperative evaluations. Apart from the use of a soft-threaded type of interference screw in the ST and the ST/G groups (Fig 2), the only variable that differed between the study groups was the type of graft. All the other factors with an important impact on the final outcome, such as the skill of the surgeons, the fixation technique, and the rehabilitation protocol, were identical. One potential weakness of the present study is that we did not use the manual-maximum KT-1000 test. It is possible that this test would have revealed a difference in laxity between the study groups especially since two thirds of the patients were male.

We were able to verify our hypothesis and, therefore, conclude that at the 2- to 3-year follow-up, both 3-strand ST grafts and 4-strand ST/G grafts produced results that were just as good as those produced by BPTB grafts in terms of functional parameters and laxity. Furthermore, donor-site problems presenting as knee-walking problems were significantly less common after using ST and ST/G grafts.

Acknowledgment: The authors thank Catarina Kartus for the illustrations.

REFERENCES

1. Beynon BD, Johnson RJ, Fleming BC, Kannus P, Kaplan M, Samani J, Renström P. Anterior cruciate ligament replacement: Comparison of bone-patellar tendon-bone grafts with two-strand hamstring grafts. A prospective, randomized study. *J Bone Joint Surg Am* 2002;84:1503-1513.
2. Eriksson E. Reconstruction of the anterior cruciate ligament. *Orthop Clin North Am* 1976;7:167-179.
3. Campell WC. Reconstruction of the ligaments of the knee. *Am J Surg* 1939;43:473-480.
4. Kartus J, Magnusson L, Stener S, Brandsson S, Eriksson BI, Karlsson J. Complications following arthroscopic anterior cruciate ligament reconstruction. A 2-5-year follow-up of 604 patients with special emphasis on anterior knee pain. *Knee Surg Sports Traumatol Arthrosc* 1999;7:2-8.
5. Aglietti P, Buzzi R, Zaccherotti G, De Biase P. Patellar tendon versus doubled semitendinosus and gracilis tendons for anterior cruciate ligament reconstruction. *Am J Sports Med* 1994;22:211-217.
6. Ejhered L, Kartus J, Sernert N, Köhler K, Karlsson J. Patellar tendon or semitendinosus tendon autografts for anterior cruciate ligament reconstruction? A prospective randomized study with a two-year follow-up. *Am J Sports Med* 2003;31:19-25.
7. Eriksson K, Anderberg P, Hamberg P, Lofgren AC, Bredenberg M, Westman I, Wredmark T. A comparison of quadruple semitendinosus and patellar tendon grafts in reconstruction of the anterior cruciate ligament. *J Bone Joint Surg Br* 2001;83:348-354.
8. Fu FH, Schulte KR. Anterior cruciate ligament surgery 1996. State of the art? *Clin Orthop* 1996;325:19-24.
9. Järvinen M, Natri A, Lehto M, Kannus P. Reconstruction of chronic anterior cruciate ligament insufficiency in athletes using a bone-patellar tendon-bone autograft. A two-year follow up study. *Int Orthop* 1995;19:1-6.
10. Buss DD, Warren RF, Wickiewicz TL, Galinat BJ, Panariello R. Arthroscopically assisted reconstruction of the anterior cruciate ligament with use of autogenous patellar-ligament grafts. Results after twenty-four to forty-two months. *J Bone Joint Surg Am* 1993;75:1346-1355.
11. Roos H, Karlsson J. Anterior cruciate ligament instability and reconstruction. Review of current trends in treatment. *Scand J Med Sci Sports* 1998;8:426-431.
12. Järvela T, Kannus P, Järvinen M. Anterior knee pain 7 years after an anterior cruciate ligament reconstruction with a bone-patellar tendon-bone autograft. *Scand J Med Sci Sports* 2000;10:221-227.
13. Shelbourne KD, Trumper RV. Preventing anterior knee pain after anterior cruciate ligament reconstruction. *Am J Sports Med* 1997;25:41-47.
14. Marder RA, Raskind JR, Carroll M. Prospective evaluation of arthroscopically assisted anterior cruciate ligament reconstruction. Patellar tendon versus semitendinosus and gracilis tendons. *Am J Sports Med* 1991;19:478-484.
15. Jansson KA, Linko E, Sandelin J, Harilainen A. A prospective randomized study of patellar versus hamstring tendon autografts for anterior cruciate ligament reconstruction. *Am J Sports Med* 2003;31:12-18.
16. Shaieb MD, Kan DM, Chang SK, Marumoto JM, Richardson AB. A prospective randomized comparison of patellar tendon versus semitendinosus and gracilis tendon autografts for anterior cruciate ligament reconstruction. *Am J Sports Med* 2002;30:214-220.
17. Aune AK, Holm I, Risberg MA, Jensen HK, Steen H. Four-strand hamstring tendon autograft compared with patellar tendon-bone autograft for anterior cruciate ligament reconstruction. A randomized study with two-year follow-up. *Am J Sports Med* 2001;29:722-728.
18. Anderson AF, Snyder RB, Lipscomb AB Jr. Anterior cruciate ligament reconstruction. A prospective randomized study of three surgical methods. *Am J Sports Med* 2001;29:272-279.
19. Noyes FR, Butler DL, Grood ES, Zernicke RF, Hefzy MS. Biomechanical analysis of human ligament grafts used in knee-ligament repairs and reconstructions. *J Bone Joint Surg Am* 1984;66:344-352.
20. Rosenberg TD. Endoscopic technique for ACL reconstruction with Pro-Trac tibial guide. Mansfield, MA: Acufex Microsurgical Inc, 1991.
21. Kurosaka M, Yoshiya S, Andrich JT. A biomechanical comparison of different surgical techniques of graft fixation in anterior cruciate ligament reconstruction. *Am J Sports Med* 1987;15:225-229.
22. Brandsson S, Faxén E, Eriksson BI, Kälebo P, Sward L, Lundin O, Karlsson J. Closing patellar tendon defects after anterior cruciate ligament reconstruction: Absence of any benefit. *Knee Surg Sports Traumatol Arthrosc* 1998;6:82-87.
23. Boszotta H, Prunner K. Refilling of removal defects: Impact on extensor mechanism complaints after use of a bone-tendon-bone graft for anterior cruciate ligament reconstruction. *Arthroscopy* 2000;16:160-164.
24. Corry IS, Webb JM, Clingeleffer AJ, Pinczewski LA. Arthroscopic reconstruction of the anterior cruciate ligament. A comparison of patellar tendon autograft and four-strand hamstring tendon autograft. *Am J Sports Med* 1999;27:444-454.
25. Daniel DM, Malcom LL, Losse G, Stone ML, Sachs R, Burks R. Instrumented measurement of anterior laxity of the knee. *J Bone Joint Surg Am* 1985;67:720-726.
26. Tegner Y, Lysholm J. Rating systems in the evaluation of knee ligament injuries. *Clin Orthop* 1985;198:43-49.
27. Hefti F, Müller W, Jakob RP, Stäubli HU. Evaluation of knee ligament injuries with the IKDC form. *Knee Surg Sports Traumatol Arthrosc* 1993;1:226-234.

28. Höher J, Bach T, Munster A, Bouillon B, Tiling T. Does the mode of data collection change results in a subjective knee score? Self-administration versus interview. *Am J Sports Med* 1997;25:642-647.
29. Kartus J, Stener S, Lindahl S, Engström B, Eriksson BI, Karlsson J. Factors affecting donor-site morbidity after anterior cruciate ligament reconstruction using bone-patellar tendon-bone autografts. *Knee Surg Sports Traumatol Arthrosc* 1997;5:222-228.
30. Sernert N. Evaluation of patients with anterior cruciate ligament ruptures, before and after reconstruction with special emphasis on the use of the KT-1000 arthrometer. Göteborg, Sweden: University of Göteborg; 2002. Thesis.
31. Tegner Y, Lysholm J, Lysholm M, Gillquist J. A performance test to monitor rehabilitation and evaluate anterior cruciate ligament injuries. *Am J Sports Med* 1986;14:156-159.
32. Shelbourne KD, Nitz P. Accelerated rehabilitation after anterior cruciate ligament reconstruction. *Am J Sports Med* 1990; 18:292-299.
33. Brandsson S, Faxén E, Kartus J, Eriksson BI, Karlsson J. Is a knee brace advantageous after anterior cruciate ligament surgery? A prospective, randomized study with a two-year follow-up. *Scand J Med Sci Sports* 2001;11:110-114.
34. Howell SM, Taylor MA. Brace-free rehabilitation, with early return to activity, for knees reconstructed with a double-looped semitendinosus and gracilis graft. *J Bone Joint Surg Am* 1996; 78:814-825.
35. Kartus J, Stener S, Köhler K, Sernert N, Eriksson BI, Karlsson J. Is bracing after anterior cruciate ligament reconstruction necessary? A 2-year follow-up of 78 consecutive patients rehabilitated with or without a brace. *Knee Surg Sports Traumatol Arthrosc* 1997;5:157-161.
36. Frank CB, Jackson DW. The science of reconstruction of the anterior cruciate ligament. *J Bone Joint Surg Am* 1997;79: 1556-1576.
37. Shino K, Pflaster DS. Comparison of eccentric and concentric screw placement for hamstring graft fixation in the tibial tunnel. *Knee Surg Sports Traumatol Arthrosc* 2000;8:73-75.
38. Pinczewski LA, Clingeleffer AJ, Otto DD, Bonar SF, Corry IS. Integration of hamstring tendon graft with bone in reconstruction of the anterior cruciate ligament. *Arthroscopy* 1997;13: 641-643.
39. Eriksson K, Kindblom LG, Wredmark T. Semitendinosus tendon graft ingrowth in tibial tunnel following ACL reconstruction: A histological study of 2 patients with different types of early graft failure. *Acta Orthop Scand* 2000;71:275-279.
40. Ballock RT, Woo SL, Lyon RM, Hollis JM, Akeson WH. Use of patellar tendon autograft for anterior cruciate ligament reconstruction in the rabbit: A long-term histologic and biomechanical study. *J Orthop Res* 1989;7:474-485.
41. Freedman KB, D'Amato MJ, Nedeff DD, Kaz A, Bach BR Jr. Arthroscopic anterior cruciate ligament reconstruction: A metaanalysis comparing patellar tendon and hamstring tendon autografts. *Am J Sports Med* 2003;31:2-11.
42. Yunes M, Richmond JC, Engels EA, Pinczewski LA. Patellar versus hamstring tendons in anterior cruciate ligament reconstruction: A meta-analysis. *Arthroscopy* 2001;17:248-257.
43. O'Neill DB. Arthroscopically assisted reconstruction of the anterior cruciate ligament. A prospective randomized analysis of three techniques. *J Bone Joint Surg Am* 1996;78:803-813.
44. Pinczewski LA, Deehan DJ, Salmon LJ, Russell VJ, Clingeleffer A. A five-year comparison of patellar tendon versus four-strand hamstring tendon autograft for arthroscopic reconstruction of the anterior cruciate ligament. *Am J Sports Med* 2002; 30:523-536.
45. Kartus J, Ejerhed L, Sernert N, Brandsson S, Karlsson J. Comparison of traditional and subcutaneous patellar tendon harvest. A prospective study of donor site-related problems after anterior cruciate ligament reconstruction using different graft harvesting techniques. *Am J Sports Med* 2000;28:328-335.
46. Kartus J, Movin T, Karlsson J. Donor-site morbidity and anterior knee problems after anterior cruciate ligament reconstruction using autografts. *Arthroscopy* 2001;17:971-980.